TSA Mock 3

Problem Solving

The longest, most sustained past TSA Problem Solving questions, from all 16 papers (2008–2023)

Questions	Total marks	Time limit
25	25	45 minutes

Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Every question is a real past TSA question. The year and number in brackets show where it comes from, e.g. **7 (2019 Q32)**.

What is in this paper

The 25 Problem Solving questions judged to keep a strong candidate thinking hard for longest, none of them in Mock 1 or Mock 2. Each needs several minutes of sustained reasoning, far more than the 1¾ minutes a question gets on average, so expect not to finish: the paper trains stamina and choosing which questions to leave.

How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to 25 questions: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes). These are hard questions, so expect to run short of time.

Score: _____ / 25

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1 (2022 Q42)

In my currency, there are six types of coin in circulation:

200p 100p 50p 20p 10p 5p

I start with two coins. I visit a shop and purchase two bars of chocolate and pay using the two coins. My change consists of two coins.

The next day I re-visit the shop and purchase another one bar of chocolate with that change. My new change, again, consists of two coins.

The six coins I possess during these two days are all different. I only have two coins at any particular time. The price of a bar of chocolate does not change.

What is the price of a bar of chocolate?

A 15p

B 25p

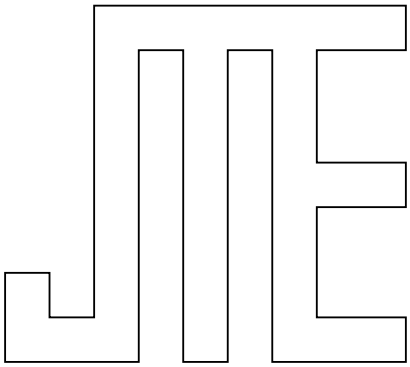
C 30p

D 55p

E 60p

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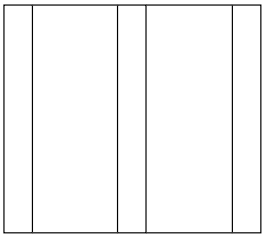
2 (2011 Q19) This is the view from above of the large three-dimensional logo situated in the forecourt of the head office of Justin Thyme Enterprises:



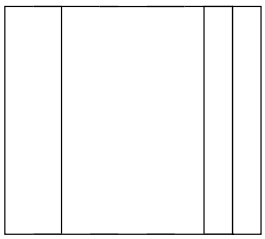
It was sculpted from a single cube of marble, and has a uniform cross-section from top to bottom.

Which of the following is **not** a possible side view of the logo?

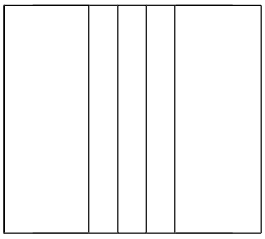
A



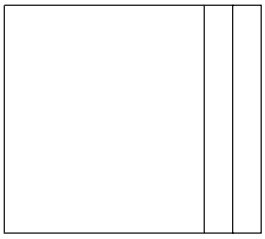
B



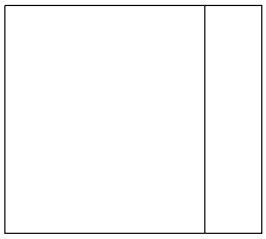
C



D



E



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3 (2012 Q37) The Ruritanian Go Kart Drivers' Championship is decided over a series of 9 races. For each event the winner is allocated 10 points, second gets 8 points and third gets 5 points. All other finishers are given 1 point. The championship leaders' placings are shown below and the drivers all finished in each race.

Running Points Total and Placings										
	Damon		Nigel		Michael		David		James	
Circuit	**Total**	**Pl.**	**Total**	**Pl.**	**Total**	**Pl.**	**Total**	**Pl.**	**Total**	**Pl.**
Ragues	10	1	5	3	8	2	1		1	
Kolm	11		6		16	2	11	1	6	3
Ostrov	19	2	11	3	17		12		16	1
Vostok	29	1	19	2	18		13		21	3
Billard	39	1	27	2	23	3	14		22	
Menz	47	2	28		24		24	1	27	3
Fondra	48		38	1	32	2	25		32	3
Hahn	49		43	3	40	2	35	1	33	
Limberg	50		51	2	41		45	1	38	3

Which was the earliest race after which one of the leaders knew he could not win the Drivers' Championship?

A Billard

B Menz

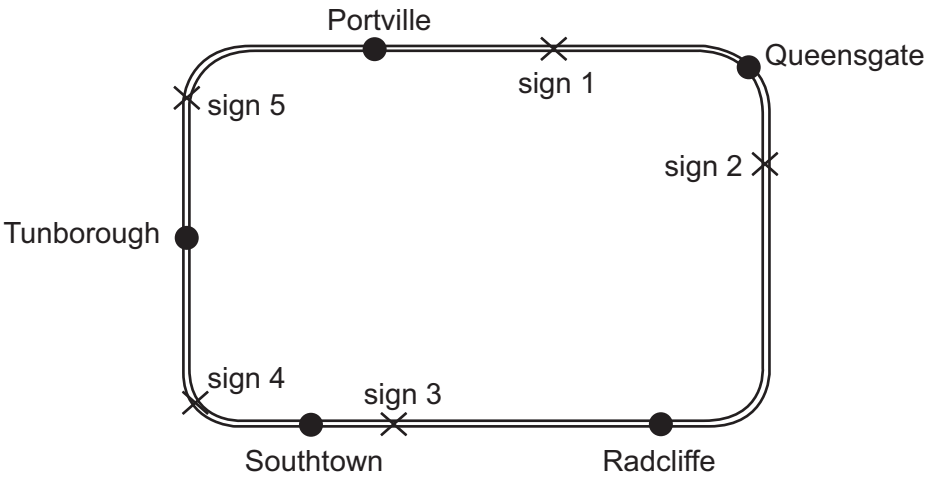
C Fondra

D Hahn

E Limberg

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4 (2022 Q50) There are five towns at various positions along a major ring road that encircles a city. Traffic travelling clockwise around the ring road is provided with information in the form of five 'distance' signs at various points along the road as shown.



The signs display the following information (all distances in km):

sign 1

Queensgate	10
Radcliffe	32
Southtown	49
Tunborough	64
Portville	88

sign 2

Radcliffe	17
Southtown	34
Tunborough	49
Portville	73
Queensgate	92

sign 3

Southtown	4
Tunborough	19
Portville	43
Queensgate	62
Radcliffe	84

sign 4

Tunborough	9
Portville	31
Queensgate	52
Radcliffe	74
Southtown	91

sign 5

Portville	13
Queensgate	32
Radcliffe	54
Southtown	71
Tunborough	86

One of the signs contains an error in one of the distances indicated.

Which one of the signs contains the error?

A sign 1

B sign 2

C sign 3

D sign 4

E sign 5

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5 (2019 Q30) Alex and Sue are playing a game with a set of nine cards, numbered from 1 to 9. A 3 × 3 grid of cards is dealt onto the table. The cards for the next round have just been dealt and are as shown below.

5	1	8
3	6	2
9	7	4

The first player chooses two cards and scores the difference between the two numbers. The second player then chooses two cards from the seven that are left and scores the average of the two numbers. The highest score wins the round. Both players must choose cards that are either horizontally or vertically next to each other in the grid.

For this round of the game Alex has to decide whether to play first or second. She wants to make sure that she wins the round. If she has more than one possible winning move then she wants to achieve the highest score possible.

What should Alex choose to do?

A play first and choose the 1 and 8

B play first and choose the 3 and 9

C play first and choose the 6 and 7

D play first and choose the 7 and 9

E play second

6 (2008 Q37) Peter needs to separate his sheep into 16 square pens. Each side of a pen is formed by a free-standing barrier known as a hurdle. Each pen must be individually accessible on at least one side so that sheep can be penned or released without a risk of the others escaping.

What is the smallest number of hurdles that Peter must take out to the fields in order to accomplish his task?

A 40
B 42
C 48
D 49
E 64

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7 (2022 Q19) A youth club is open six evenings every week. There are nine volunteers who run the club. Each evening three of the volunteers are on duty.

When the rota is drawn up for any week, the following conditions must be met:

- Each volunteer has two evenings of duty, but never two consecutive evenings.
- No two volunteers are on duty together twice in the same week.

Next week's rota has just been published, as follows:

Monday	*Tuesday*	*Wednesday*	*Thursday*	*Friday*	*Saturday*
Mel	Eve	Jan	Mel	Eve	Fay
Rod	Fay	Leo	Pat	Jan	Leo
Sam	Tim	Sam	Tim	Rod	Pat

Fay wants to swap her Tuesday duty with one of the others to be on duty on Monday or Thursday instead.

Who is the only volunteer on the rota for Monday or Thursday that Fay could swap with if the conditions described above are still to be met?

A Mel

B Pat

C Rod

D Sam

E Tim

8 (2020 Q24) I had planned my day trip to the stately home very carefully, but then the bus was late. I won't arrive until 11:30 am and will have to leave by 4:15 pm. There were five things that I had wanted to do and I had written these down in the table below:

activity	*length of time*	*start times*
visit the museum	55 minutes	11 am or 2 pm
visit the maze	55 minutes	every half hour (at quarter past and quarter to the hour)
take the guided tour	80 minutes	1 pm or 3 pm
walk in the grounds	75 minutes	any time
have lunch in the café	25 minutes	between 12:30 pm and 2:00 pm

Which activity should I miss out so that I can still do the other four?

A visit the museum

B visit the maze

C take the guided tour

D walk in the grounds

E have lunch in the café

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9 (2018 Q42) In the main draw of a lottery, six of the balls (which are numbered from 1 to 49) are selected at random, then the numbers chosen are rearranged and displayed in ascending numerical order.

For instance: 3 17 20 29 34 45

In one draw recently, I noticed that the six numbers were made up of a total of ten digits, all different. The lowest number on this occasion was 1 and the highest number was 49, so the range of the six numbers was 48, the greatest possible range for the lottery main draw.

What is the smallest possible range when the six numbers in the main draw of the lottery have a total of ten digits, all different?

A 21

B 23

C 27

D 31

E 32

10 (2019 Q25) Sandi is planning a trip to the old city of Pordson. She wants to stay for four nights in order to visit all of the attractions, and one of the nights must be the Saturday so that she can attend a special music concert.

The table below shows the prices per night and details of the deals for the five main hotels in Pordson.

Hotel	*Mon*	*Tues*	*Weds*	*Thurs*	*Fri*	*Sat*	*Sun*		*Deal*
Grand	€60	€70	€70	€70	€50	€60	€60		3F2
Majestic	€40	€90	€70	€70	€40	€90	€40		SFSHP
Palace	€60	€90	€90	€70	€50	€90	€80		50%
Palms	€60	€80	€80	€50	€40	€60	€50		4FP3
Plaza	€70	€90	€90	€60	€50	€70	€50		CBK

Deals:
- 3F2 Stay 2 nights and get the 3rd night free
- SFSHP Stay Friday and get the Saturday night half price.
- 50% 50% discount on all stays of 3 or more nights.
- 4FP3 Stay 4 nights and get the cheapest night free.
- CBK €50 discount on stays of 3 or more nights.

What is the minimum that Sandi will need to spend on a hotel bill for her stay in Pordson?

A €110

B €120

C €140

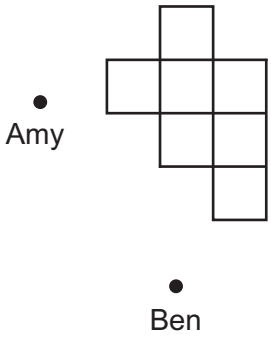
D €145

E €160

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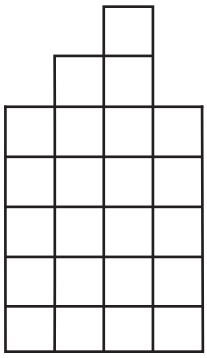
11 (2018 Q26) A number of identical boxes are piled up, waiting to be loaded into a delivery van. Amy and Ben are about to load them into the van.

The appearance of the pile of boxes and the positions of Amy and Ben at present are shown below:



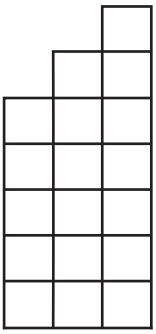
[view from above]

Amy's view of the pile is:

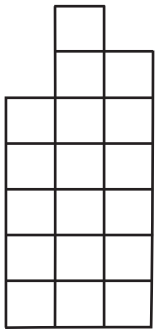


Which one of the following could **not** be Ben's view of the pile?

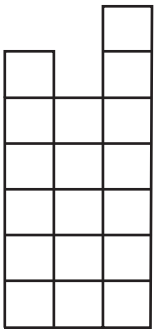
A



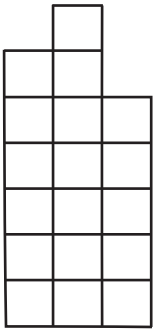
B



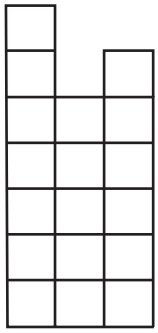
C



D



E



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12 (2023 Q18) Five friends play a word game called Wordsmith. The objective is to score as many as points as possible, using the letters of a randomly chosen starting word to create new words. Each letter used in a word scores one point, and the player with the most points after five rounds is the winner.

The word chosen for this particular game was 'RAINBOW'. In the first four rounds, the words selected by the five players were as follows:

Georgina	*Riaz*	*Bobby*	*Vandy*	*Sonny*
BRAIN	BROWN	ROBIN	BRAWN	WORN
WARN	BARON	BORN	IRON	BOAR
BRAN	RAIN	WAR	BROW	BAN
NOW	WON	AN	NO	OR

The words given in the final round by the five players were: 'ROW', 'BAR', 'AIR', 'ON' and 'I' (not in order).

Which one of the following is **not** a possible difference between the final scores of the players with the most and least points?

A 1

B 2

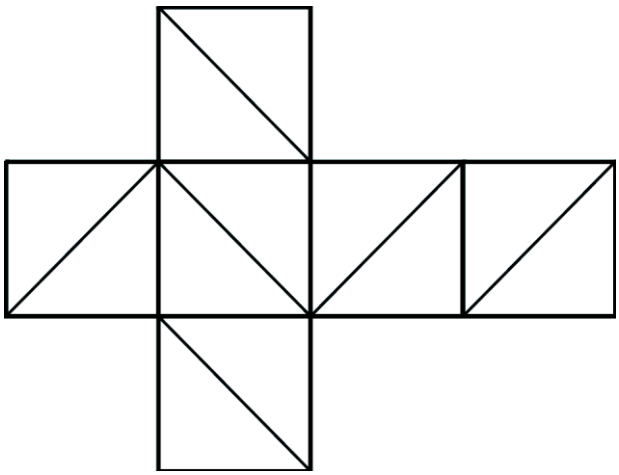
C 3

D 4

E 5

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13 The following net is folded up to make a cube.
(2017 Q26)



At how many corners of the cube do exactly two of the diagonal lines on the faces meet?

A 2

B 3

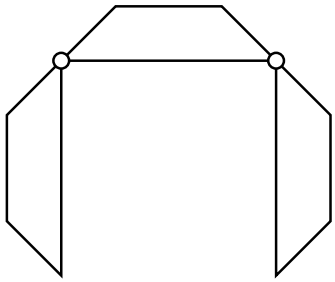
C 4

D 5

E 6

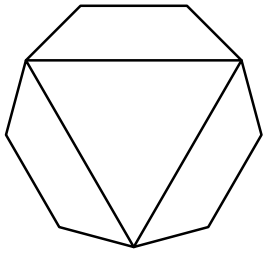
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14 (2009 Q8) The diagram below shows a rather unusual conference table which may take up a range of configurations. The three pieces are hinged together at the points shown by the circles:

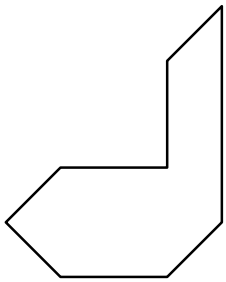


Which of the following shapes could **NOT** be made from this table?

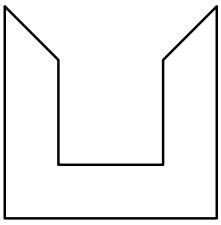
A



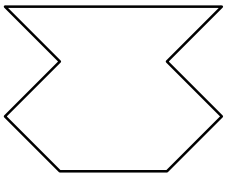
B



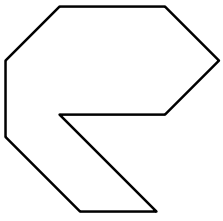
C



D

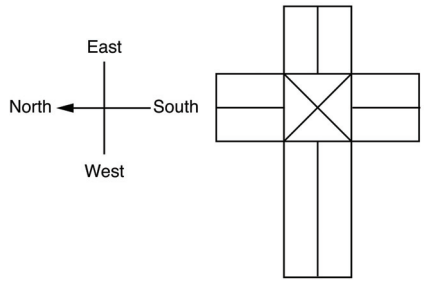


E



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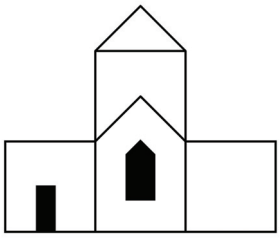
15 The diagram below shows a view of a church from above:
(2013 Q44)



There are three windows facing south, two windows facing north and a stained glass window facing east. The only doors are one in the westernmost end and one in a north-facing wall.

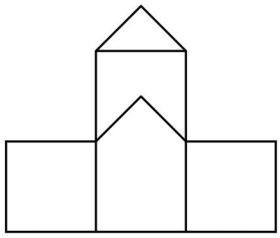
Which one of the following gives a possible view of the church?

A



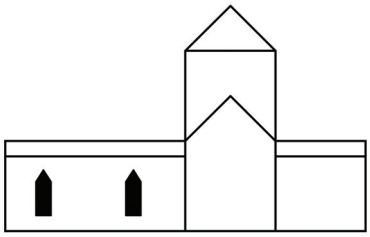
View from East

B



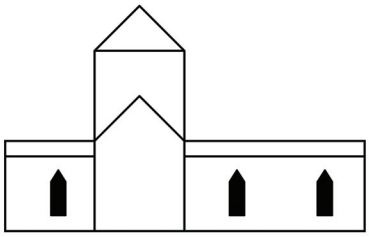
View from West

C



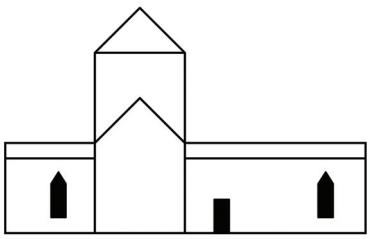
View from South

D



View from South

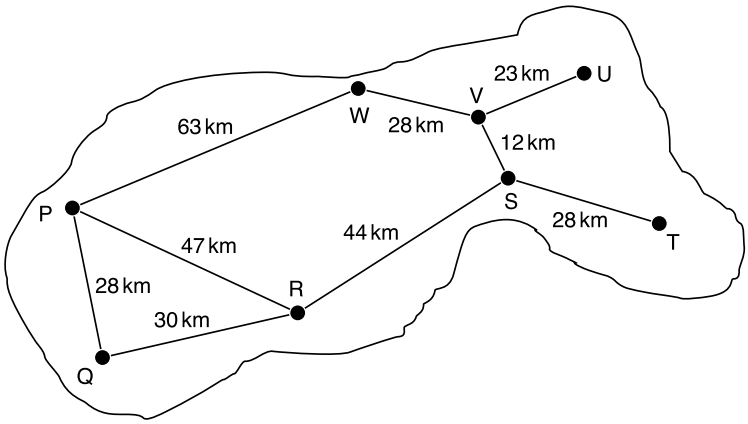
E



View from North

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16 (2009 Q26) This is a map of the island of Metallica showing the eight main towns (labelled P - W) and the roads that connect them, together with a chart giving the shortest distance (in km) by road between each pair of towns:



Copper	Gold	Iron	Lead	Nickel	Silver	Tin	Zinc
74							
28	91						
86	12	91					
102	28	119	40				
91	40	63	28	68			
109	35	114	23	63	51		
30	44	47	56	72	84	79	

Which of the towns on the map is Nickel?

A Q

B R

C S

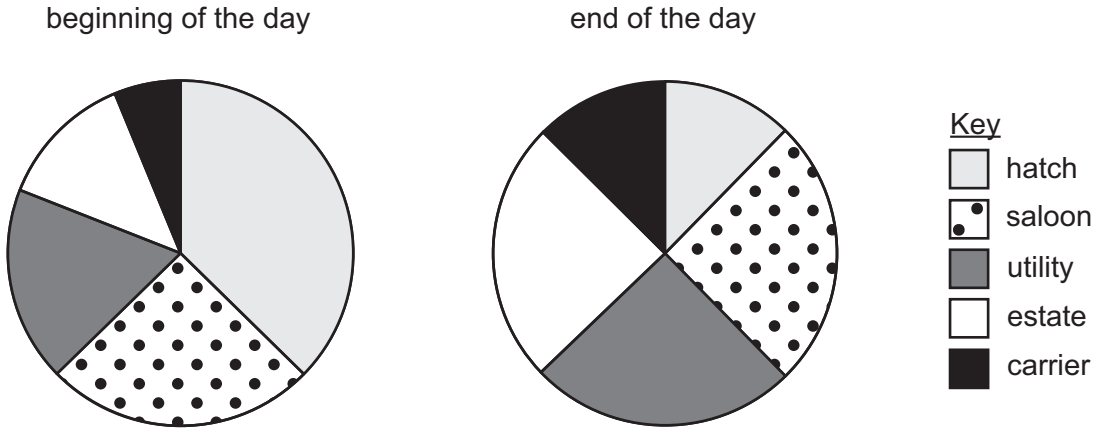
D T

E W

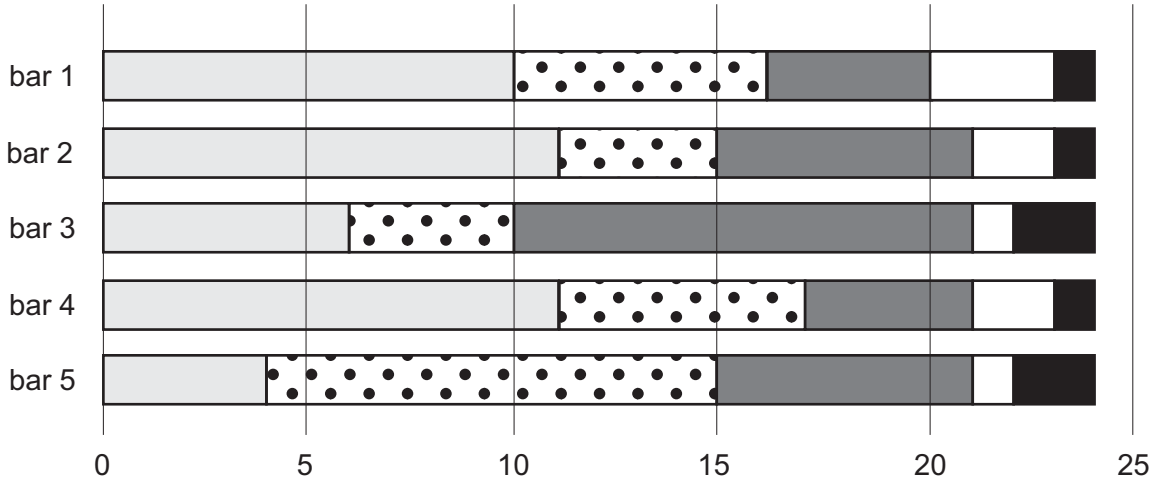
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17 (2022 Q38) At the beginning of the day, a showroom had 32 vehicles for sale. The vehicles were advertised in five categories: hatchback, saloon, utility, estate car and people carrier. By the end of the day, 24 vehicles had been sold.

The first chart shows a comparison of the numbers of vehicles in each category in the showroom at the beginning of the day, and the second chart shows a comparison of the numbers in the same categories in the showroom at the end of the day.



Which one of the following shows a comparison of the numbers of cars in each category that were sold?



A bar 1

B bar 2

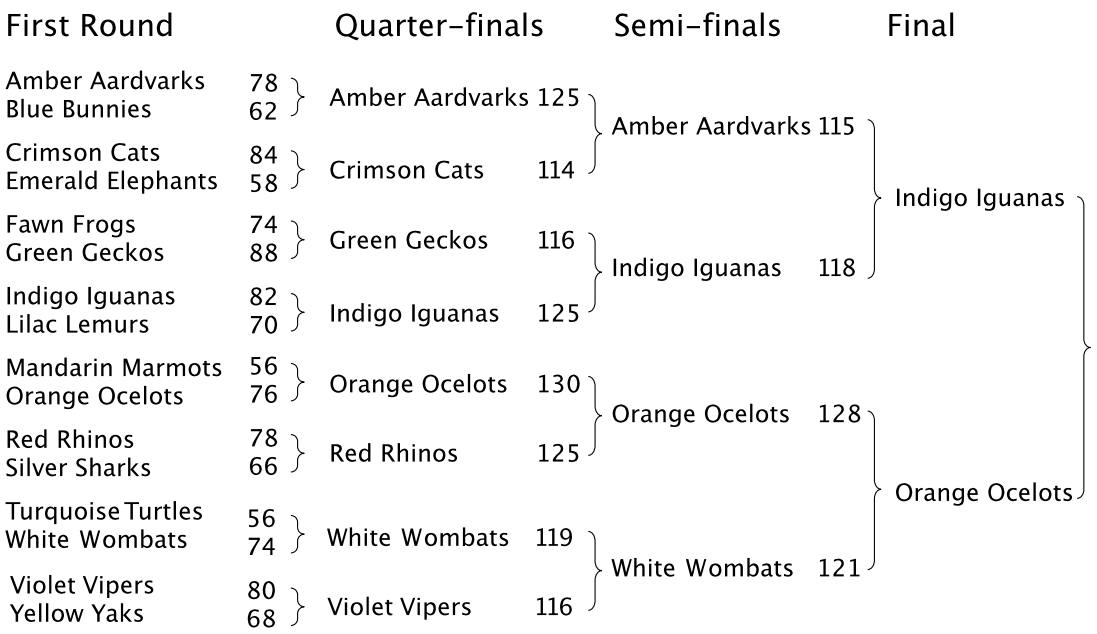
C bar 3

D bar 4

E bar 5

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18 (2010 Q43) This is the draw for this year's end of season knockout trophy (and the results so far) competed for by sixteen teams that take part in a quiz league:



Although teams are drawn against each other, all participating teams in each round answer the same set of 50 questions, worth 2 points each.

In the first round, each 'match' is won straightforwardly by the team with the higher number of points. This team carries half of these points through to the next round; then in subsequent rounds winners carry through half of their 'new' points from the previous round. For example, the Amber Aardvarks carried 39 points through to their quarter-final, then in the quarter-final itself they added 86, and so carried 43 through to their semi-final.

How many more points (at least) must the Indigo Iguanas score than the Orange Ocelots in the final to win the trophy?

A 4

B 6

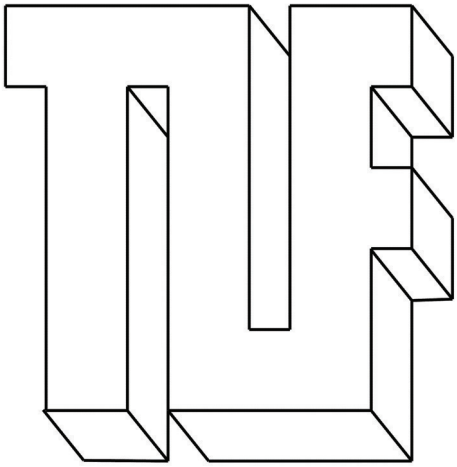
C 8

D 10

E 12

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19 (2014 Q8) Below is Thomas Leslie Fuller's brass paperweight, which shows his initials.



Which one of the following is **not** a side view of Thomas' paperweight when it is placed flat on a table (either side up)?

A



B



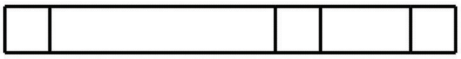
C



D

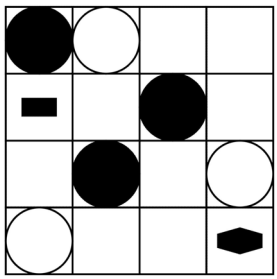


E



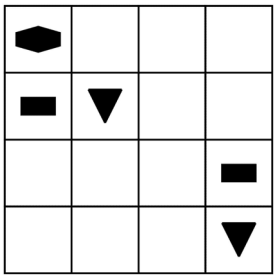
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20 (2012 Q14) In a game, circular counters are moved around a board which is a 4 x 4 grid. There are symbols on some of the squares on the board. At one point in the game, the board looks like this (all the circles represent counters which completely cover any symbol on the square that they occupy):

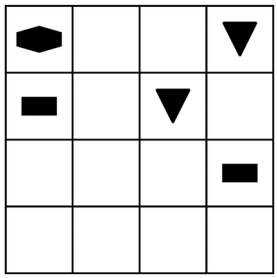


Which one of the following diagrams could show the layout of the board (not necessarily viewed in the same direction as above)?

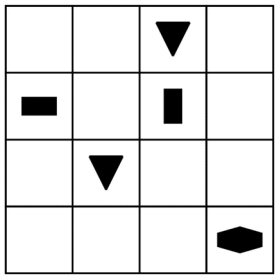
A



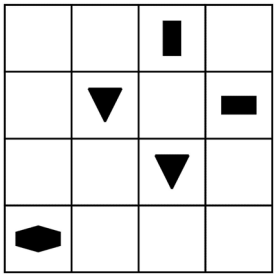
B



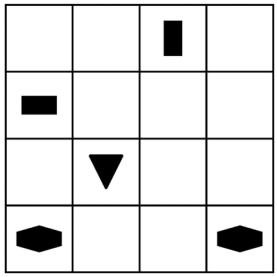
C



D

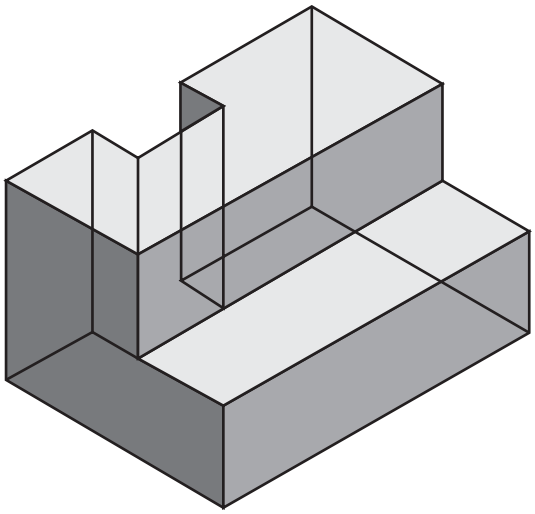


E



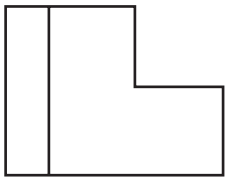
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21 (2019 Q20) The see-through glass block below can be viewed from six sides, including from above and below.

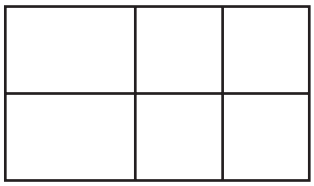


Which one of the following is **not** a possible view?

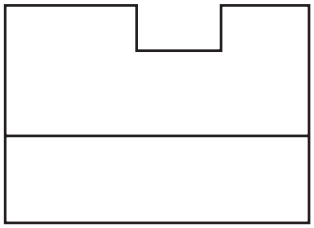
A



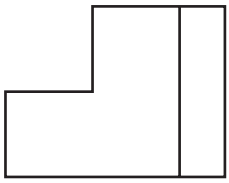
B



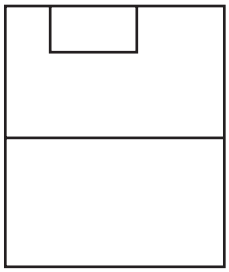
C



D

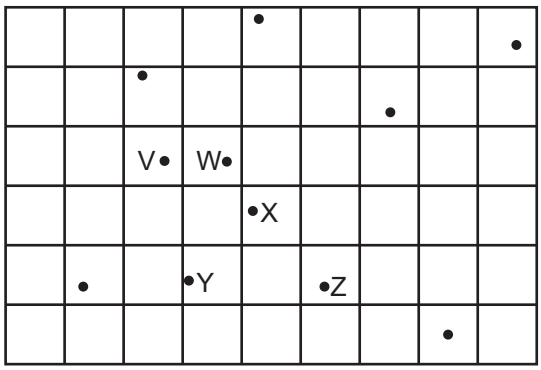


E

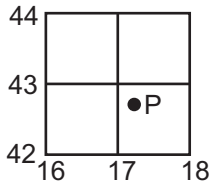


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22 The diagram below shows part of a map of Ruralia.
(2018 Q38)



A group of students are on an expedition and must meet their teacher at Longwood, grid reference 787634. Unfortunately, they have been given a version of the map with the numbers missing from the gridlines, and the place names have been omitted too. They have a list of grid references for some places on the map, and are trying to work out which point to head for in order to be sure of meeting their teacher.



They have a set of instructions showing how the grid references work:

The grid reference of point P is 172427. Go to vertical line to the west of the point, read the number 17. Estimate how many tenths across the square the point appears at, hence 172. Go to horizontal line to the south of point P, read the number 42, estimate how many tenths up the square the point is, hence 427. This gives 172427 as the complete grid reference number.

Here is the list of the grid references / place names the students have been given.

Hilltop	763613	Riverside	781614
Sunnybank	804613	Topfield	777634
High Moor	792626	Longwood	787634
Highscar	773649	Stakefell	793658
West Pasture	815642	Lower Green	837654
Little Tarn	825605		

Using this information to help you, which of the places marked on the map is Longwood?

A V

B W

C X

D Y

E Z

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23 (2018 Q30) Recently, packets of Amblers crisps have had ‘money-off’ coupons inside them. Some coupons are worth 9p, some are worth 14p and some are worth 20p.

George has been collecting these coupons. He has more 14p coupons than 9p coupons, and more 9p coupons than 20p coupons. The total value of all of his coupons is exactly £1.50.

How many coupons has George collected?

A 9

B 10

C 11

D 12

E 13

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24 (2015 Q43) Hester is going for a job interview. She needs to travel by railway between Birmingham and Derby, but she needs to stop off in Tamworth for an hour on the way out to see her former teacher for some coaching. When at Derby, she needs an hour and a half to go to the prospective employer to take an aptitude test. She can fix her appointments for any time in the day. Naturally, she wants to make the journey in the most efficient way so that it takes the least possible time.

This is the railway timetable:

Out:

Station								
Birmingham	09:00	10:00	11:00	12:00	13:00	13:30	14:30	15:30
Tamworth	09:30	10:28	11:30	12:45	-	13:45	14:45	15:50
Derby	09:50	10:40	11:50	13:05	13:35	14:00	15:05	16:00

Return:

Station								
Derby	10:40	11:10	12:15	13:10	14:00	15:20	16:10	17:00
Tamworth	11:05	-	12:35	13:45	-	15:45	16:35	-
Birmingham	11:50	11:55	13:10	14:15	15:00	16:12	16:55	18:15

Which would be the best trains to catch from Birmingham and Derby?

A Leaving Birmingham 09:00; leaving Derby 12:15

B Leaving Birmingham 10:00; leaving Derby 14:00

C Leaving Birmingham 11:00; leaving Derby 15:20

D Leaving Birmingham 12:00; leaving Derby 16:10

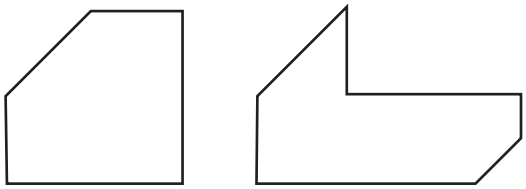
E Leaving Birmingham 13:30; leaving Derby 17:00

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25 (2018 Q32)

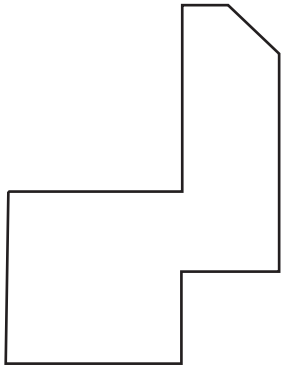
Connem Ltd. has recently adopted a new company logo.

A large plastic version of the logo has just been delivered to the office, to be attached to the outside of the building. Unfortunately, in transit, the plastic has broken into two pieces, as shown below:

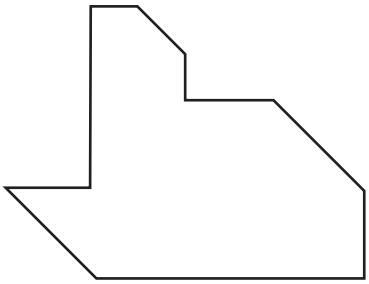


Which one of the following could **not** be Connem's new logo?

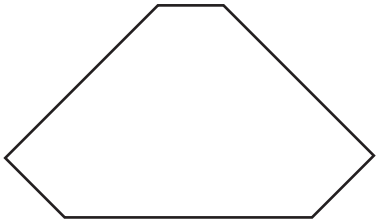
A



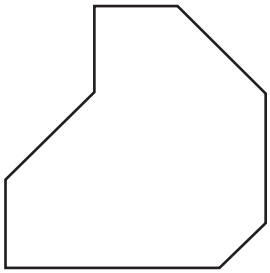
B



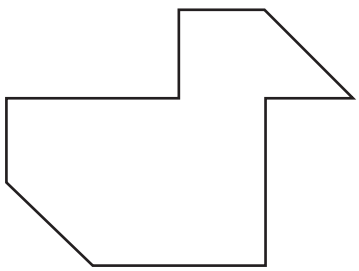
C



D



E



END OF TEST

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